

Deep Neural Networks for Doubly Robust Estimation with Nonprobability Survey Samples

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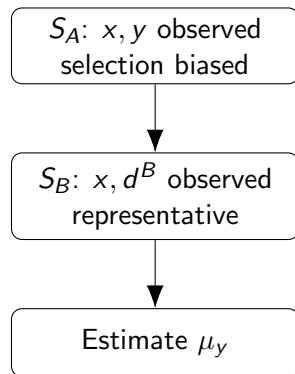
Based on : Dai, Luo, Lou, Wang & Lu (2026)

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Motivation: integrating two imperfect data sources

- **Nonprobability sample** S_A : rich outcome y , but unknown selection mechanism.
- **Probability sample** S_B : design weights and representative covariates x , but often no y .
- Goal: estimate the finite-population mean

$$\mu_y = \frac{1}{N} \sum_{i=1}^N y_i.$$



Formal setup and notation

- Finite population $U = \{1, \dots, N\}$; target parameter: $\mu_y = \frac{1}{N} \sum_{i=1}^N y_i$.
- **Nonprobability sample** S_A , size n_A : observe (x_i, y_i) . Selection indicator $R_i = I(i \in S_A)$, with $\pi_i^A = P_q(R_i = 1 \mid x_i, y_i)$.
- **Probability sample** S_B , size n_B : observe (x_i, d_i^B) , where $d_i^B = 1/\pi_i^B$ and $\pi_i^B = P(i \in S_B)$ are the first-order inclusion probabilities.
- **Ignorability assumption**: $\pi_i^A = P(R_i = 1 \mid x_i)$ — selection depends only on observed x , not on y .

Three classical strategies

- 1 Model the outcome: $m(x_i) = E(y_i \mid x_i) \rightarrow$ predict y for units in S_B .
- 2 Model the selection: $\pi^A(x) = P(R = 1 \mid x) \rightarrow$ inversely weight units in S_A .
- 3 Combine both models (doubly robust).

Method 1: Regression / mass imputation (REG)

Procedure

- 1 Fit outcome model $m(x_i) = E(y_i | x_i)$ using S_A (e.g., OLS).
- 2 Predict $\hat{y}_i = m(x_i, \hat{\beta})$ for every unit in S_B .
- 3 Estimate the population mean by

$$\hat{\mu}_{REG} = \frac{1}{\hat{N}_B} \sum_{i \in S_B} d_i^B \hat{y}_i, \quad \hat{N}_B = \sum_{i \in S_B} d_i^B.$$

Critical assumption

$m(x)$ is correctly specified. If $E(y | x)$ contains interactions, thresholds, or nonlinearities and a linear model is used, $\hat{\mu}_{REG}$ is inconsistent — no amount of design weighting can salvage a misspecified outcome model.

Method 2: Inverse probability weighting (IPW)

Procedure

- 1 Pool S_A and S_B ; define $R_i = I(i \in S_A)$.
- 2 Fit propensity model $\pi^A(x) = P(R = 1 | x)$, typically logistic: $\text{logit}[\pi^A(x)] = \alpha_0 + x^\top \alpha$.
- 3 Weight each unit in S_A inversely to its selection probability:

$$\hat{\mu}_{IPW} = \frac{1}{\hat{N}_A} \sum_{i \in S_A} \frac{1}{\hat{\pi}_i^A} y_i, \quad \hat{N}_A = \sum_{i \in S_A} \frac{1}{\hat{\pi}_i^A}.$$

Critical assumption

$\pi^A(x)$ follows the assumed parametric form (e.g., linear logit). If the true selection mechanism involves **nonlinearities, interactions, or unknown functional forms**, the logistic model is misspecified and IPW is inconsistent.

Method 3: Doubly robust (DR) estimation

Formula

$$\hat{\mu}_{DR} = \frac{1}{\hat{N}_A} \sum_{i \in S_A} \frac{1}{\hat{\pi}_i^A} \{y_i - m(x_i, \hat{\beta})\} + \frac{1}{\hat{N}_B} \sum_{i \in S_B} \frac{1}{\pi_i^B} m(x_i, \hat{\beta}).$$

Double robustness property

$\hat{\mu}_{DR}$ is consistent if **either** the outcome model $m(x)$ **or** the propensity model $\pi^A(x)$ is correctly specified — not necessarily both. This is a genuine improvement over REG and IPW alone.

Remaining vulnerability

When **both** working models are misspecified — e.g., the outcome regression is linear but $E(y | x)$ is nonlinear, *and* the propensity is logistic but the true $\pi^A(x)$ is nonlinear — the DR estimator can also be inconsistent.

Parametric double robustness $\neq$ nonparametric protection.

Key idea of the paper

The paper replaces the parametric propensity-score model by a DNN model:

$$\log \frac{\pi_i^A}{1 - \pi_i^A} = g_0(x_i),$$

where g_0 is an unknown nonparametric function.



The DNN is trained using information from both the nonprobability sample and the reference probability sample.

Why deep neural networks?

- **Universal approximation:** multilayer feedforward networks can approximate broad classes of continuous functions on compact sets.
- **Curse of dimensionality:** under smoothness and structural (e.g. composite Hölder) assumptions, sparse DNNs can achieve the optimal minimax rate without suffering from the curse of dimensionality.
- **Scalability:** compared with classical nonparametric methods (e.g. kernel smoothing), DNNs scale better in moderate-to-high dimensions when the target function has suitable structure.
- **Nonlinear selection:** participation mechanisms in nonprobability samples often involve unknown interactions and nonlinearities — precisely the setting where DNNs excel.

Key insight

The parametric logit model $\text{logit}[\pi^A(x)] = \alpha_0 + x^\top \alpha$ is a **special case** of the DNN model. The DNN relaxes the linear-in-parameters restriction and lets the data determine the functional form of the selection mechanism.

Pseudo-likelihood: derivation

Step 1: Population log-likelihood (infeasible)

If x_i were observed for all $i \in U$, the log-likelihood for the participation indicators R_i would be

$$\ell(g) = \sum_{i=1}^N [R_i \log \pi_i^A + (1 - R_i) \log(1 - \pi_i^A)] = \sum_{i \in S_A} \log \frac{\pi_i^A}{1 - \pi_i^A} + \sum_{i=1}^N \log(1 - \pi_i^A).$$

Step 2: Replace population total by HT estimator

Since x_i is unavailable outside $S_A \cup S_B$, the population sum $\sum_{i=1}^N \log(1 - \pi_i^A)$ cannot be computed. The **Horvitz–Thompson estimator** uses the probability sample S_B :

$$\sum_{i=1}^N \log(1 - \pi_i^A) \approx \sum_{i \in S_B} d_i^B \log(1 - \pi_i^A), \quad d_i^B = \frac{1}{\pi_i^B}.$$

Step 3: Nonparametric logit model

Under the DNN model $\log \frac{\pi_i^A}{1-\pi_i^A} = g(x_i)$, we have $\log(1 - \pi_i^A) = -\log\{1 + \exp[g(x_i)]\}$.

The pseudo-log-likelihood simplifies to

$$\ell^*(g) = \sum_{i \in S_A} g(x_i) - \sum_{i \in S_B} d_i^B \log\{1 + \exp[g(x_i)]\}$$

- First term: uses units from the nonprobability sample S_A .
- Second term: uses the probability sample S_B as a design-weighted proxy for the population.
- A sparse ReLU DNN approximates g_0 ; ADAM minimizes $-\ell^*(g)$.

DNN architecture and training

Network structure

A $(K + 1)$ -layer feedforward DNN with width vector $\mathbf{p} = (p_0, \dots, p_K, p_{K+1})$:

- $p_0 = r$ (input: covariate dimension); $p_{K+1} = 1$ (output: scalar logit).
- K hidden layers with ReLU activation: $\sigma(u) = \max\{0, u\}$.
- Sparse network class $\mathcal{G}(K, s, \mathbf{p}, D)$: total number of non-zero parameters $\leq s$, sup-norm $\leq D$.

Training

- Loss function: $-\ell^*(g)$ (negative pseudo-log-likelihood).
- Optimizer: ADAM (adaptive moment estimation).
- After training: $\hat{\pi}_i^A = \{1 + \exp[-\hat{g}(x_i)]\}^{-1}$.

Proposed estimators

After training the DNN,

$$\hat{\pi}_i^A = \{1 + \exp[-\hat{g}(x_i)]\}^{-1}.$$

DNN-assisted IPW

$$\hat{\mu}_{DIPW} = \frac{1}{\hat{N}_A} \sum_{i \in S_A} \frac{1}{\hat{\pi}_i^A} y_i, \quad \hat{N}_A = \sum_{i \in S_A} \frac{1}{\hat{\pi}_i^A}.$$

Deep doubly robust estimator

$$\hat{\mu}_{DDR} = \frac{1}{\hat{N}_A} \sum_{i \in S_A} \frac{1}{\hat{\pi}_i^A} \{y_i - m(x_i, \hat{\beta})\} + \frac{1}{\hat{N}_B} \sum_{i \in S_B} \frac{1}{\pi_i^B} m(x_i, \hat{\beta}).$$

Assumptions: identification and design

A1: Ignorability (Missing At Random)

$\pi_i^A = P(R_i = 1 \mid x_i, y_i) = P(R_i = 1 \mid x_i)$. Selection into S_A depends only on x_i , not on y_i . Hence $E(y \mid x) = E(y \mid x, R = 1)$, so the outcome model is estimable from S_A .

A2: Positivity

$\pi_i^A > 0$ for all $i \in U$. Every unit has a nonzero chance of entering S_A . Together with A1, this is the **strong ignorability** condition (Rosenbaum & Rubin, 1983).

A3: Conditional independence

$R_i \perp R_j \mid x_i, x_j$ for $i \neq j$. Membership indicators are independent across units given covariates — rules out cluster or network dependence in the selection process.

Assumptions: DNN complexity and regularity

B1: DNN architecture

The network depth K , width p_k , and sparsity s scale as

$$K = O(\log n), \quad s = O(n\gamma_n^2 \log n), \quad \min_k p_k \asymp \max_k p_k \asymp n.$$

Deep enough to approximate g_0 , sparse enough to avoid overfitting.

B2: Smoothness of g_0

$g_0 \in \mathcal{H}(q, \beta, \mathbf{d}, \tilde{\mathbf{d}}, M)$: a composite Hölder class. Each component has smoothness β_i and intrinsic dimension $\tilde{d}_i \leq d_i$. Sparse ReLU networks approximate such g_0 at the rate γ_n .

Assumptions: regularity conditions

Design conditions (C1, C4, C5)

- **C1:** $n_A/N \rightarrow f_A \in (0, 1)$, $n_B/N \rightarrow f_B \in (0, 1)$ — samples are of comparable order.
- **C4:** HT estimators are $\sqrt{n_B}$ -consistent for population totals of x_i , y_i , $m(x_i, \beta)$, $m'(x_i, \beta)$.
- **C5:** $N\pi_i^A/n_A$ and $N\pi_i^B/n_B$ bounded away from 0 and ∞ — like simple random sampling.

Technical conditions (C2–C3, C6–C9)

- **C2–C3:** $m(x, \beta)$ twice continuously differentiable in β with bounded derivatives.
- **C6:** Finite moments of y_i^2 , $\|x_i\|^3$ bounded; propensity-score information matrix positive definite.
- **C7:** Logistic link $\sigma(z) = (1 + e^{-z})^{-1}$ has globally bounded derivatives.
- **C8:** Uniform design-consistency of HT estimators over the sparse DNN class.
- **C9:** Estimated scores truncated to $[\kappa_n, 1 - \kappa_n]$ for numerical stability.

Theorem 1: Convergence rate of the DNN estimator

Setup

g_0 belongs to a composite Hölder class $\mathcal{H}(q, \beta, \mathbf{d}, \tilde{\mathbf{d}}, M)$ where each component has smoothness β_i and intrinsic dimension $\tilde{d}_i \leq d_i$. The overall rate is $\gamma_n = \max_{i=0, \dots, q} n^{-\tilde{\beta}_i / (2\tilde{\beta}_i + \tilde{d}_i)}$ with $\tilde{\beta}_i = \beta_i \prod_{k=i+1}^q (\beta_k \wedge 1)$.

Theorem 1

Under A1–A3, B1–B2, C1, C8: $\|\hat{g} - g_0\|_{L^2([0,1]^r)} = O_p(\gamma_n \log^2 n)$.

Key insight

The rate depends on the **intrinsic dimensions** $\tilde{d}_i$, not the ambient r . When $\tilde{d}_i \ll d_i$, DNNs circumvent the curse of dimensionality.

Theorem 2: Convergence of DIPW and DDR

Theorem 2

Under A1–A3, B1–B2, C1–C9 (including correct specification of g_0):

$$|\hat{\mu}_{DIPW} - \mu_y| = O_p(\gamma_n \log^2 n), \quad |\hat{\mu}_{DDR} - \mu_y| = O_p(\gamma_n \log^2 n).$$

Interpretation

Both estimators inherit the DNN convergence rate, which depends on the composite smoothness and intrinsic dimension of g_0 , not the raw covariate count.

Setup

- $N = 20\,000$, $n_A = 500$, $n_B = 1000$, $B = 500$
- Outcome: $y_i = 2 + \sum_{j=1}^4 x_{ji} + \sigma \varepsilon_i$
- $\rho = \text{corr}(y, x^\top \beta) \in \{0.30, 0.50, 0.80\}$

True propensity (nonlinear)

$$\text{logit}(\pi_i^A) = \theta_0 + 0.05x_1x_2 + 0.1x_2^2 + 0.05x_3x_4 \\ + 0.08 \sin(0.3x_3) + 0.05 \ln(1 + x_2 + x_4)$$

TF: outcome correct, propensity wrong

Fitted logit is linear in x ; omits interactions, squares, sin, log.

FF: both models wrong

Outcome: x_4 omitted; Propensity: same linear logit.

Simulation results ($\rho = 0.50$)

TF scenario

Estimator	%RB	MSE
Naive ($\bar{y}$)	74.67	47.97
REG	<u>-0.67</u>	<u>0.25</u>
IPW	-5.87	7.47
DR	0.73	4.06
DIPW	2.10	1.70
DDR	<u>-0.46</u>	0.62

REG best (outcome correct);
DDR close behind via DNN propensity.

FF scenario

Estimator	%RB	MSE
Naive ($\bar{y}$)	74.66	47.97
REG	79.06	53.81
IPW	-5.80	7.48
DR	-24.21	10.07
DIPW	2.77	1.77
DDR	<u>-0.26</u>	<u>1.49</u>

REG bias flips (+79%);
DR worse than IPW (-24.2%);
DDR stable (-0.26%).

DDR reduces IPW's MSE by **92%** (TF) and **80%** (FF).

Empirical application

Data

- PRC 2015: $n_A = 9\,301$ (nonprob.)
- BRFSS 2015: $n_B = 441\,456$ (prob.)
- 8 covariates, 7 outcomes
(civic/political participation,
drinking)

Covariate imbalance (%)

	PRC	BRFSS
White only	82.3	75.1
Hispanic	9.3	16.5
HS or less	21.6	42.7
Bachelor's+	41.6	26.3

PRC oversamples whites & college-educated.

Selected estimates (%)

Outcome	Naive	REG	IPW	DR	DIPW	DDR
Trust neighbors	59.0	54.7	54.9	54.7	51.7	52.1
Express opinions	26.5	23.8	23.9	23.9	26.8	22.3
Voted local elections	75.0	70.0	70.3	70.1	71.0	70.1

DDR: trust -6.9pp , opinions -4.2pp $\Rightarrow$ selection bias matters for civic outcomes.

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